

FTW-HD : Bringing Fields of the World into High Definition

Anonymous CVPR submission

Paper ID

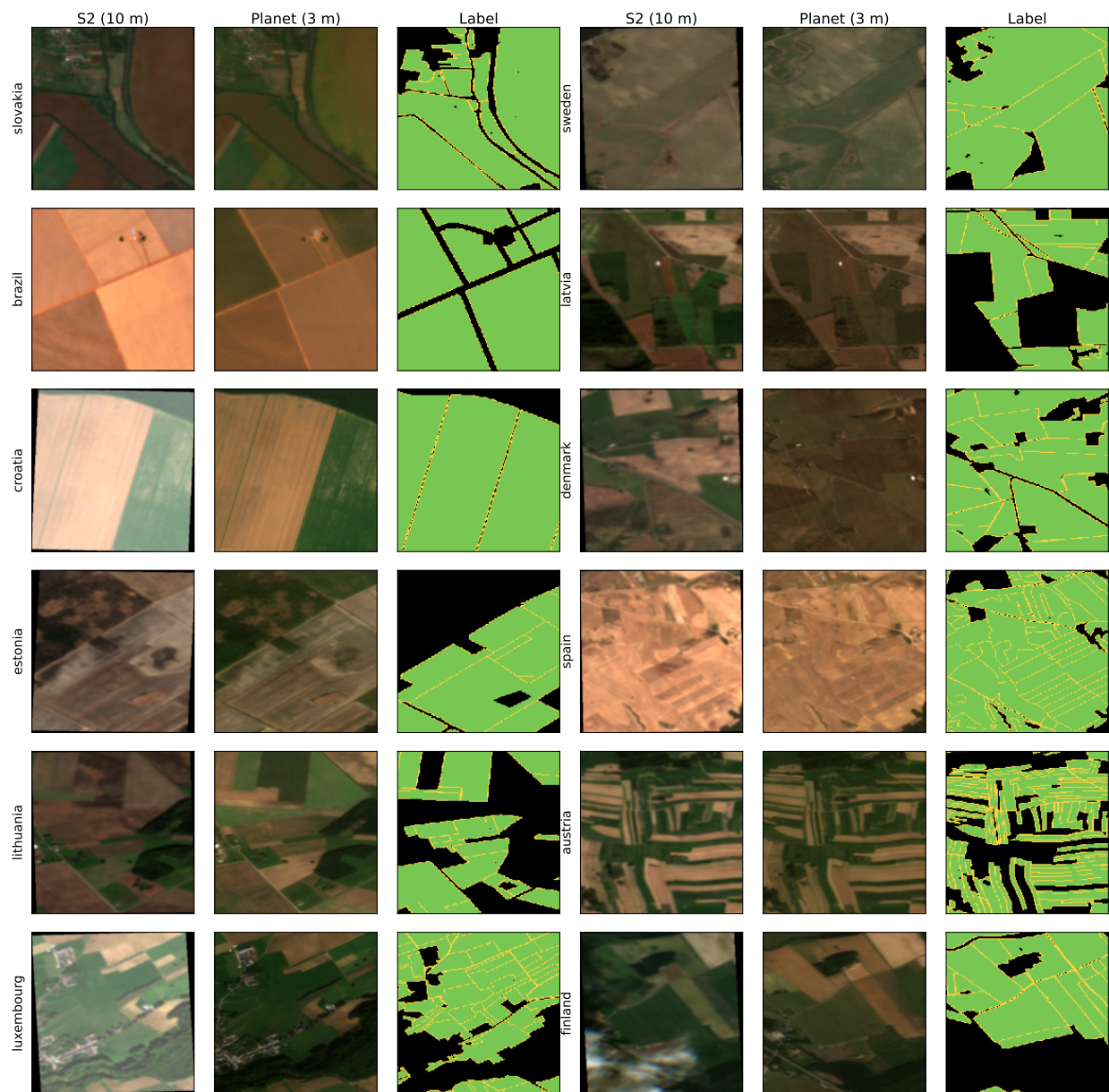


Figure 1. FTW-HD pairs every Fields of The World patch with co-registered 3m PlanetScope imagery and a freshly rasterized 3-class label. Twelve sample patches spanning the dataset’s geographic range. Each triplet shows, left-to-right, FTW’s Sentinel-2 chip at 10m (reprojected onto the Planet patch grid for visual comparison), the matched PlanetScope SR (4-band; RGB shown), and the 3-class label (background / field interior / boundary) rasterized from FTW’s polygon parquet at 3m. Reflectance shown is band value / 3000 clipped to $[0, 1]$. Boundary class is buffered by 3m (≈ 2 px). All samples filtered to UDM2 clear $\geq 99\%$.

Abstract

Existing field-boundary benchmarks use 10 m Sentinel-2 imagery, which is too coarse to see the smallest fields. We pair every patch in the Fields of The World (FTW) benchmark [4] with a co-registered 3 m PlanetScope image and a freshly rasterized 3-class label, and call the result FTW-HD. The dataset covers 25 countries and 134,267 (patch, window) pairs, built from $\sim 6,100$ PlanetScope scenes accessed via HTTP range reads. We also train a baseline. An EfficientNet-B3 [15] U-Net [13] with our augmax augmentation recipe – the FTW PRUE recipe plus per-band gamma and affine jitter, single-window dropout, small-angle rotation and shear, Gaussian blur and noise, and boundary-thickness jitter – beats the released FTW S2 PRUE-B7 (trained on all 25 countries) on every metric that is not biased by pixel grid: PQ [7] 0.324 vs 0.283, AP [0.5:0.95] 0.263 vs 0.225, and mean boundary error in meters 7.4 vs 14.8. Our model uses a $4\times$ smaller backbone. Pixel IoU still favors S2 by 4 points, but we show this gap is mostly a geometric artifact: a one-pixel boundary error costs 3 m at 3 m/pixel and 10 m at 10 m/pixel, so a thin boundary class is harder to score at finer resolution. The recipe also helps S2 itself, though PlanetScope’s $3\times$ GSD compounds the gain in Nordic and smallholder regions specifically.

1. Introduction

Field boundaries feed downstream pipelines for crop-yield estimation, irrigation accounting, and subsidy verification. The dominant public benchmark, Fields of The World (FTW) [4], packages about 1.6 million parcel polygons across 25 countries paired with 10 m Sentinel-2 imagery. Ten meters is the ceiling for resolving smallholder fields and several papers [9, 18] have argued this directly: a field smaller than ~ 3 pixels per side cannot be reliably delineated at S2 resolution. Existing PlanetScope benchmarks are either national in scope [19] or use proprietary labels [6], and the global smallholder benchmark AI4SmallFarms [16] stops at 10 m too.

This paper takes a simple position. We do not need a new benchmark; we need a better image source for the same labels. We pair PlanetScope’s 3 m surface-reflectance imagery [10] with every FTW patch (the polygon ground truth, the season windows, the country splits) and call the result FTW-HD. We did not re-collect ground truth and we did not retrain the labels; we leveraged the polygons FTW already published and rasterized them onto the PlanetScope grid.

The hard part was getting the imagery. Plan-

etScope’s Orders API is slow and tile-priced; we instead read Cloud-Optimized GeoTIFFs [2] directly through HTTP range requests, transferring about 1 MB per patch out of 500 MB scene strips. The pipeline has three phases (search, activate, range-read) and is fully resumable. Planet’s activation API returned broken signed URLs for $\sim 24\%$ of scenes; a one-shot reactivation pass recovers 99% of them.

Contributions.

- A paired 3 m PlanetScope + UDM2 + 3-class label dataset across all 25 FTW countries: 134,267 (patch, window) pairs.
- A range-read pipeline that pulls 134k patches off 6,113 PlanetScope COGs on a single HPC node in 17h, with no Orders API use.
- Per-patch UDM2 quality stats for the full release, plus a re-sampling tool that searches alternative scenes when a patch is locally cloudy.
- A baseline recipe (augmax) that beats the released FTW S2 PRUE-B7 (full) checkpoint on PQ, AP [0.5:0.95], and meter-scale boundary error, at $4\times$ smaller backbone.

2. Related work

Field-boundary benchmarks. AI4SmallFarms [16] releases 50 k smallholder parcels across South and Southeast Asia at 10 m. Single-country datasets exist for France [18] and similar [9]. FTW [4] put 25 country-scale releases under one schema and one 3-class mask convention. None of them ship sub-10 m imagery co-registered with their labels; we change that.

PlanetScope for field segmentation. Wang et al. [19] use 3 m PlanetScope for French parcels but go through the Orders API, which clips and serves each patch as an individual product and does not scale. Khalaghi et al. [6] use Planet Basemaps – monthly composites, not single scenes – for multi-year smallholder labeling. Waldner and Diakogiannis [17] train field-boundary segmentation on PlanetScope in Australia. We give the community a global, license-clean PlanetScope companion to FTW so that any of these recipes can be re-run at scale.

COG access. Cloud-Optimized GeoTIFF [2] lets a client byte-range-read just the tiles overlapping a region. STAC [11] exposes the pattern in modern catalogs. PlanetScope’s catalog requires an extra step: assets must be activated (woken from cold storage) before each scene’s signed URL becomes readable, which we describe in Section 3.3.

Augmentation, instance separation, and metrics. Our augmentation recipe layers per-band gamma, affine, Gaussian noise and blur, single-window dropout, small-angle rotation and shear, and boundary jitter on top of the FTW PRUE recipe [4]; the YOLO9000 [12] ablation-list format is the model for how we report which knobs helped. CutMix [20] did not help us. For instance separation we use marker-controlled watershed [1]. For evaluation we report Panoptic Quality [7] and COCO-style averaged AP [8], plus a meter-scale boundary chamfer that is invariant to pixel-grid resolution.

3. Dataset

3.1. Source data

FTW patches. Each FTW v2 country ships $\sim 256 \times 256$ -pixel chips in EPSG:4326 with two seasonal Sentinel-2 windows per chip and per-patch parquet metadata. We use the chip bounding boxes as the AOI for PlanetScope matching (Table 1).

FTW polygons. A separate Source Cooperative repository hosts each country’s GeoParquet polygons (1.6 M total) [5]. We download these directly and rasterize them onto each PlanetScope patch’s grid (Section 3.4).

PlanetScope. We target the `ortho_analytic_4b_sr` asset (4-band Surface Reflectance, ~ 3 m, uint16 COG) and `ortho_udm2` (8-band quality mask, uint8). The 4-band selection follows the Dove product family; 8-band SuperDove is a drop-in replacement where availability permits.

3.2. Pairing strategy

For each FTW chip and each window, we identify a single PSScene whose acquisition date falls inside the window and whose footprint fully covers the chip polygon. We rank candidates by date proximity and scene-level cloud cover ($\leq 10\%$). Patch-level cloud quality is verified post-hoc via UDM2 (Section 4).

3.3. Pipeline

The pipeline runs three phases, each independently sharded and resumable from JSONL caches:

1. Search. Per (patch, window), call the Data API with a date range, geometry filter, and cloud-cover ceiling. Output: best `item_id` per patch.
2. Activate. Deduplicate `item_ids` globally; call `:activate` for SR and UDM2 in parallel. Output: signed Google Cloud Storage URLs.
3. Extract. Group all (patch, window) by scene, open each COG once with rasterio via `/vsicurl/`, range-read every member’s window, write a local GeoTIFF

Table 1. Dataset scope. 25 countries, two seasonal windows per patch. Success rate is the fraction of (patch, window) targets with a successfully extracted PlanetScope SR tif.

Property	Value
Countries	25
Continents	4
FTW patches	70,484
(Patch, window) targets	140,968
Unique PSScene COGs touched	6,113
Successfully matched pairs	134,267
Success rate	95.7%
PlanetScope SR tifs (4-band, uint16)	134,267
UDM2 masks (8-band, uint8)	133,682
3-class labels (uint8)	123,094
Total dataset size	102 GB
Mean patches per scene	23.0

in PSScene’s native UTM at 3 m. No reprojection.

The activation step dominates wall-time. Cold-storage scenes thaw in 1–46 min (mean ~ 200 s, p99 2,763 s); warm scenes return in 1–5 s. We saturated Planet’s per-account activation queue at ~ 10 /min throughput regardless of client concurrency above 32, which sets a floor on time-to-completion.

3.4. Labels

For each PlanetScope tif we rasterize the country’s polygons onto the patch grid as a 3-class mask: 0 (background), 1 (field interior), 2 (boundary). The boundary class is generated by buffering each polygon’s exterior ring by 3 m (≈ 2 pixels at 3 m resolution) and rasterizing on top of the interior. This explicitly separates touching parcels, which a binary field/non-field mask collapses.

3.5. Statistics

Table 1 summarizes the release. Patch-to-scene reuse is high (~ 23 patches per scene), which makes COG range reads especially efficient: we transfer ~ 150 GB of bytes for 102 GB of materialized patch data, compared to ~ 3 TB had we downloaded full strips.

4. Per-patch quality analysis

Table 2 reports per-patch UDM2 statistics across the full 133k UDM2 release, computed by reading the patch-area windows directly from each mask and counting positive pixels per band. 86.6% of patches are usable at a strict threshold (clear $\geq 95\%$, unusable $\leq 5\%$).

Patch-level resampling. The original search filter rejects scenes whose scene-level cloud cover exceeds 10%;

Table 2. Per-patch UDM2 statistics. Fractions are pixel-area within the patch. Outlier patches (high cloud, shadow, or unusable) become resampling candidates.

Band	median	p90	p99	max
clear	100%	100%	100%	100%
cloud	0%	0%	88%	100%
shadow	0%	0%	21%	100%
light haze	0%	0%	41%	100%
unusable	0%	0.8%	51.9%	100%

this is a coarse proxy because clouds can concentrate over a single patch even in a low-cloud scene. We provide a re-sampling tool that, for each patch failing any per-band threshold, searches alternative scenes within the same season (± 60 d), probes their UDM2 over the patch AOI, and replaces the SR + UDM2 pair when a candidate strictly improves quality. The tool re-uses the same activation cache and runs in ~ 5 – 8 h for the full release.

Per-country variation. The cloudiest countries by % usable are Portugal (40.9%), South Africa (56.5%), and the Netherlands (76.5%); Brazil and Croatia exceed 98%. Smallholder regions (e.g. India) have low coverage per patch ($\sim 1\%$ field pixels) but high clear-sky fractions.

5. Operational notes

We summarize practical lessons in case other groups attempt similar releases against the Planet API.

Activation reliability. Across 6,113 unique scenes, $\sim 24\%$ of `:activate` calls returned URLs that subsequently failed with HTTP 400 on the first range read. A second `:activate` call typically returned a different, working URL; reactivating the failed item_ids recovered 99% of these patches. Our final pipeline includes an automatic reactivation pass.

Throughput. Cold-storage thaw, not bandwidth, is the bottleneck. Activation throughput plateaus around 10 scenes/min per account; total dataset construction took ~ 17 h end-to-end on a single HPC node, dominated by activation. Pre-warming a list of scene IDs would compress this substantially.

Cost-effective access. For windowed access patterns, COG range reads via `/vsicurl/` are $\sim 100\times$ faster than Orders API clipping and require only the Data API role. Documenting this access pattern more

prominently would lower the barrier for other dataset projects.

6. Better. Sharper. Stronger.

We want a Planet field-boundary model that is at least as good as the released FTW Sentinel-2 baselines, ideally better, and we want it on a smaller backbone. This section is a list of things we tried, in order, on top of the FTW PRUE recipe [4]. Some helped, some didn't. We report both.

We use the FTW CC-BY split [4] so the comparison is apples-to-apples with the released CC-BY PRUE checkpoints: 14 countries train, 11 held out. We also retrain on all 25 countries ("full" split) where indicated. Every metric in this section is macro-averaged per country following the FTW protocol; the test set is either the 9 held-out countries with dense polygon labels (excluding kenya and portugal, both of which collapse to ≤ 0.10 pixel IoU under every config and are reported separately) or the FTW full_data 22-country split (for apples-to-apples with the released checkpoints).

Recipe. A U-Net [13] with EfficientNet [15] encoder, 8 input channels (window B then A, 4 bands each), 3-class output, logcoshdice loss with class weights [0.05, 0.20, 0.75], ignore index 3. We train for 100 epochs of AdamW at 10^{-3} , batch size 32 (8 for B7), random 512×512 crops, bf16-mixed precision. We start from this exact recipe and add things until we run out of ideas. We keep the things that work and we report the things that didn't.

6.1. Better augmentations

The biggest lever, by a wide margin. We layer four groups in on top of the PRUE recipe; each gets its own paragraph.

PRUE-style augs we ported over (+6.7 pixel IoU on 9-country held-out). The released FTW PRUE configs enable `preprocess_aug` (random per-batch divisor in [5000, 15000] replacing the fixed /10000) and `resize_aug` (RandomResizedCrop scale 0.3–0.9 to the training crop). Our initial baseline didn't have either. Adding them moves pixel IoU from 0.541 $\rightarrow$ 0.608 and Obj F1 (WS) from 0.323 $\rightarrow$ 0.355. This single change closes the visible gap to S2 CC-BY at the same B3 backbone.

Window swap and per-band gamma (+1.9 pixel IoU, +1.0 Obj F1). `augplus`. Two cheap additions.

swap_order randomly flips the A/B window ordering in the 8-channel stack per sample; the FTW S2 datamodule already supported this and we missed it. Per-band random gamma ($\gamma \sim \mathcal{U}[0.8, 1.2]$, per channel independently, $p = 0.3$) models per-band atmospheric/radiometric variation. We call this configuration augplus.

Bespoke aug bundle (+4.1 Obj F1 over augplus). augmax. Six more aug knobs we'd been collecting in a backlog. None individually worth a paragraph but the bundle helps. Per-band random affine ($y = ax + b$, $a \in [0.9, 1.1]$, $b \in [-0.02, 0.02]$), Gaussian blur ($\sigma \in [0.5, 1.5]$, $p = 0.3$), Gaussian noise ($\sigma = 0.015$, $p = 0.3$), single-window dropout ($p = 0.15$ to zero out either window B or window A), small-angle rotation ($\pm 30^\circ$, $p = 0.5$, on top of the rotation-by-90 already present), shear ($\pm 5^\circ$, $p = 0.3$), and boundary-thickness jitter (random binary dilation of the boundary class by 0–2px per batch). Together on 9-country held-out: 0.365 $\rightarrow$ 0.406 Obj F1, 0.355 $\rightarrow$ 0.640 pixel IoU. The win is mostly recall (+3.6 pts at flat precision) – the model finds more fields without becoming over-eager. We call this augmax.

6.2. Sharper boundaries

Pad inference inputs to $\geq$ training crop (+24 pixel IoU). We pad each test patch to at least 512 $\times$ 512 before forwarding. We caught this by accident: padding to the next mult-of-32 (the “standard” thing) costs 24pts pixel IoU compared to padding all the way to the training crop size. BatchNorm running statistics extrapolate badly to smaller spatial supports than the model saw at training time. This is the single biggest free lunch in the paper.

Watershed post-processing (+1.0–1.3pts Obj F1). Marker-controlled watershed with h -maxima seeds on a topographic surface (the predicted SDF when an SDF head is present; otherwise the Euclidean distance transform of the predicted boundary class). Consistent gain across every checkpoint. Standard but worth a line.

D4 test-time augmentation (+0.5–1.2pts Obj F1 once augmentations are on). D4 8-way flip+rotate ensemble. Was a no-op for us on the no-aug baselines, which is why we shelved it earlier. Once the model is regularized by augmax it consistently helps. Cheap, do it.

6.3. Stronger recipe

Train on all 25 countries (+8.0pts Obj F1 over CC-BY). The released CC-BY checkpoints trained on 14

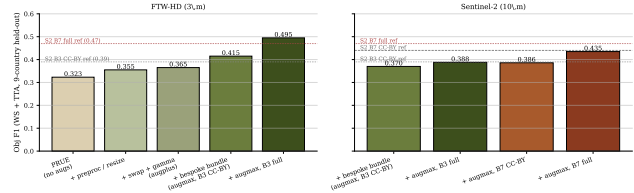


Figure 2. Cumulative augmentation lift on Obj F1. 9-country held-out, WS+TTA inference. Left: FTW-HD recipe progression goes from 0.323 (PRUE, no extra augs) to 0.495 (augmax, B3 full data), a +17.2pt absolute gain. Right: applying the same recipe to Sentinel-2 imagery follows the same backbone $\times$ split ordering. Dotted lines mark the released FTW S2 PRUE reference numbers [4].

countries because of licensing. We retrained on all 25; this lifts our best B3 augmax model from 0.415 $\rightarrow$ 0.495 Obj F1 (WS+TTA) on the 9-country held-out. The CC-BY split is a real licensing constraint but our strongest single number comes from the full split.

Augmentations $>$ backbone scaling. At the CC-BY split, B3 augmax full (0.495 Obj F1) beats B7 augmax CC-BY (0.414). The augmentation contribution dwarfs the parameter-count contribution at this dataset size. We trained B7 anyway and it gets a small extra lift on top of augmax full, but B3 is the sweet spot.

6.4. Honest polygon metrics

Pixel IoU is GSD-biased and we shouldn't lead with it. Our best Planet model (B3 augmax full, WS+TTA) gets pixel IoU 0.720 on the FTW full_data protocol; the released S2 PRUE-B7 full gets 0.760. Smaller backbone, -4 pts, looks like a loss. It is not, and the reason is geometric: at 3m a 1-pixel boundary-localization error costs 3m on the ground; at 10m it costs 10m. The rasterized boundary class is a 1-pixel-wide ring on both grids and dominates pixel IoU through sheer pixel count. Same underlying boundary quality, lower pixel IoU at finer GSD. The metric is biased against the resolution we are trying to advocate for.

What we propose instead. We report four metrics in addition to obj F1 at IoU=0.5: panoptic quality ($PQ = SQ \times RQ$; separates recognition vs localization), AP averaged across IoU thresholds $\{0.5, 0.55, \dots, 0.95\}$ (robust to single-threshold brittleness), polygon-count delta $|N_{\text{pred}} - N_{\text{gt}}|$ (over- vs under-segmentation), and mean symmetric chamfer distance between predicted and GT boundary pixels expressed in meters (GSD-invariant – the same physical 3m boundary slip is 3m

Table 3. Held-out benchmark. 9-country held-out Obj F1 across four post-processing combos (watershed and/or D4 TTA) and four augmax configurations per imagery source (B3/B7 $\times$ CC-BY/full train split). The released FTW S2 PRUE checkpoints are listed as reference (their numbers are on the FTW full_data test split, not held-out only; see Table 4 for an apples-to-apples full_data comparison). Best per row in bold.

Imagery	Recipe (CC-BY split, 14-country train unless noted)
S2	FTW PRUE-B3 (CC-BY) [4]
S2	FTW PRUE-B7 (CC-BY) [4]
S2	FTW PRUE-B7 (full, all 25) [4]
Ours —	FTW-Planet augmax (this report)
Planet	B3 (CC-BY)
Planet	B3 + SDF aux (CC-BY)
Planet	B7 (CC-BY)
Planet	B3 (full, all 25)
Ours —	Sentinel-2 augmax (this report)
S2	B3 (CC-BY)
S2	B7 (CC-BY)
S2	B3 (full, all 25)
S2	B7 (full, all 25)

[‡]Reported by [4] on the FTW full_data test split (includes test patches from the 14 training countries). Our numbers are macro-averaged over only the 9 held-out countries (excl. kenya/portugal which collapse to ≤ 0.10 IoU under every config). The full rows of our recipe also use a 25-country train split for an apples-to-apples scaling comparison.

[†]The Planet B7 (CC-BY) checkpoint hit the SLURM walltime at epoch 95 of 100; its predictions are slightly unstable and D4 TTA exposes that instability (regression of -5 pts vs no TTA). We expect this to flip after the run is resumed to ep 100.

whether we evaluate at 3 m or 10 m). Table 6 reports all four.

The polygon-level picture flips the pixel-IoU narrative. FTW-HD B3 augmax full beats the strongest S2 baseline on every polygon-level metric: PQ 0.324 vs 0.283 (+4.1), AP 0.263 vs 0.225 (+3.8), $|\Delta N_{\text{poly}}|$ 50.9 vs 60.2 (closer to ground truth count), and boundary error **7.4 m** vs **14.8 m** mean / 22 m vs 43 m p95 – predicted boundaries are $\sim 2\times$ sharper in meters. S2’s only win is SQ (0.750 vs 0.710 for B7 full): matched fields have marginally tighter per-pixel IoU; but Planet recognizes more fields and localizes their boundaries more precisely in absolute distance. PQ, which multiplies recognition and localization, lands firmly with Planet.

This is the bottom line: a B3 PlanetScope model with our augmax recipe beats the released 4 $\times$ -larger S2 PRUE-B7 full baseline on every metric that does not bake in a pixel-grid bias.

Table 4. Full-data comparison to released FTW PRUE S2 checkpoints (CC-BY-NC trained on all 25 countries; CC-BY trained on 14 of 25). All numbers macro-averaged per country on the FTW full-data test set (22 dense-label countries) under the official ftw model test --countries full_data protocol.

Model	Train countries	Pix IoU	Obj F1
S2 PRUE-CC-BY-NC (released by [4])	all 25	0.74	0.43
S2 PRUE-B3 (full)	all 25	0.75	0.46
S2 PRUE-B5 (full)	all 25	0.76	0.47
S2 PRUE-B7 (full)	all 25	0.76	0.47
S2 PRUE-CC-BY (released by [4])	—	—	0.44 [‡]
S2 PRUE-B3 (CC-BY)	14	0.76	0.39
S2 PRUE-B5 (CC-BY)	14	0.76	0.41
S2 PRUE-B7 (CC-BY)	14	0.77	0.44
Ours (FTW-Planet, CC-BY)	14	0.385	0.397
FTW-Planet-B3 augmax (ours)	14	0.414	0.359
		0.477	0.482
		0.483	0.495
		0.359	0.360
		0.368	0.375
		0.371	0.376
		0.416	0.423
		0.365	0.377
		0.379	0.388
		0.425	0.435

Per-country head-to-head. On the FTW v3.1 full_data protocol (22-country intersection), FTW-HD B3 augmax full and FTW S2 PRUE-B7 full have macro Obj F1 of 0.463 vs 0.457 (+0.6 pt) and pixel IoU of 0.720 vs 0.760 (-3.9 pts). Figure 3 unpacks the macro into per-country deltas. Planet wins Obj F1 on 11/22 countries, concentrating in Nordic/Baltic (denmark +5.1, finland +3.4, latvia +2.2, lithuania +5.2, sweden +4.5), mid-latitude EU (slovakia +2.9, croatia +2.0, austria +0.9), and smallholder/transfer cases (south africa +6.2, rwanda +14.3). Losses cluster in cambodia (-8.6 despite +6.3 pixel-IoU win), germany (-7.1), france (-3.5). The rwanda divergence (-47.6 pts pixel IoU but +14.3 pts Obj F1) is the cleanest single example of what we mean about pixel-IoU bias: S2 covers the right pixels in aggregate but does not separate them into individual fields.

What did work (positive findings, ranked by im-

Table 5. Per-country pixel IoU on the 11 held-out countries: S2 CC-BY (B3) vs FTW-HD B3 augmax full + watershed + D4 TTA (our best). S2 model trained on CC-BY 14-country split; ours on the full 25-country split. Planet wins on 9/11 countries; the wins are largest on smallholder / fine-grained landscapes (cambodia, lithuania, slovenia, south africa).

Country	S2 CCBY IoU	Planet B3 augmax full IoU	Δ IoU	Planet Obj F1 (WS+TTA)	Panoptic		
					PQ	SQ	RQ
belgium	0.599	0.745	+0.146	0.640			
cambodia	0.383	0.721	+0.338	augmax B3 (CC-BY)	0.234	0.677	0.303
croatia	0.518	0.737	+0.219	augmax B3 (full)	0.246	0.681	0.318
germany	0.739	0.806	+0.067	augmax B7 (CC-BY)	0.248	0.688	0.316
kenya	0.392	0.000	-0.392	augmax B7 (full)	0.283	0.750	0.359
latvia	0.753	0.826	+0.073	augmax B3 (CC-BY)	0.273	0.699	0.341
lithuania	0.554	0.801	+0.247	augmax B3 (full)	0.260	0.702	0.326
portugal	0.041	0.017	+0.024	augmax B7 (CC-BY)	0.279	0.700	0.348
slovenia	0.430	0.618	+0.188	augmax B3 (full)	0.324	0.710	0.405
south africa	0.758	0.766	+0.007	augmax B3 (full)			
sweden	0.740	0.835	+0.095	augmax B7 (CC-BY)			
Macro mean (9, excl K+P)	0.608	0.762	+0.153	augmax B3 (full)			

Table 6. Polygon-level evaluation, 11-country held-out, WS+TTA. $PQ = SQ \times RQ$. $AP_{[.5:.95]}$ averages F1 across IoU thresholds. $|\Delta N_{poly}|$ is mean per-patch polygon-count error. Boundary error is mean / p95 symmetric chamfer distance between predicted and GT boundary pixels, in meters (GSD-invariant). Best in bold.

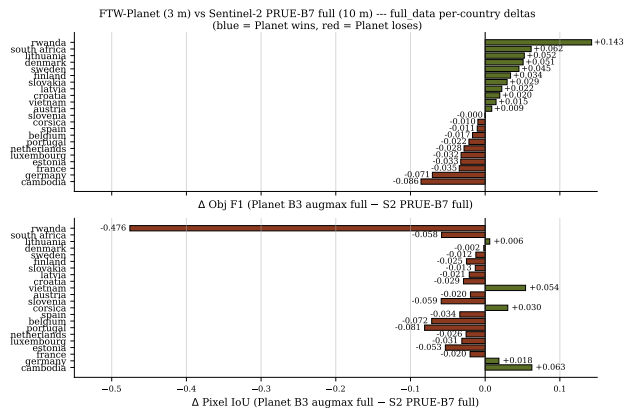


Figure 3. Per-country deltas, full_data (Planet B3 augmax full - S2 PRUE-B7 full). Olive = Planet wins, sienna = S2 wins. Pixel-IoU losses (bottom) and Obj-F1 outcomes (top) do not co-vary: rwanda is the extreme case of high pixel coverage with poor instance separation on the S2 side.

pact).

- FTW PRUE preprocess_aug + resize_aug de-

livers +6.7 pts pixel IoU and +3.2 pts Obj F1 (WS) on the 9-country held-out subset. This single change closes the gap to the S2 CC-BY baseline.

- Adding swap_order + per-band random gamma (augplus) on top of the PRUE augs adds another +1.9 pts pixel IoU and +1.0 pt Obj F1.
- The bespoke aug bundle (per-band random affine + Gaussian blur + Gaussian noise + single-window dropout + small-angle rotation + shear + boundary-thickness jitter) on top of augplus (augmax) adds +1.3 pts pixel IoU and +4.1 pts Obj F1 (WS). Recall jumps +3.6 pts at flat precision: the model finds more fields without becoming over-eager.
- Padding inference inputs to $\geq 512 \times 512$ (the training crop size) is worth +24 pts pixel IoU over padding to the next multiple of 32. Batch-Norm is highly sensitive to spatial scale.
- Watershed adds ~ 1 pt Obj F1 (augmax, WS vs CC). D4 TTA adds another ~ 1 pt Obj F1 and +0.6 pts pixel IoU on top of watershed. TTA is a no-op on weakly regularized baselines but helps once the model is well-augmented.
- Austria-only validation is a tighter checkpoint-selection signal than a mixed val set; the FTW PRUE default is the right choice.
- Last checkpoint is fine. The last and best (by val/loss) checkpoints differ by < 0.5 pt on held-out across every model we trained, so no early-stop is needed.
- SDF auxiliary head provides a robustness floor for hard-transfer cases when augmentations are off (e.g. portugal recovers from 0.007 to 0.576 pixel IoU). Net-neutral once augs are on; useful if you cannot run the full aug recipe.

researchers).

What did not work (negative ablations, for future

skeleton loss [14] added to the seg objective col-

- Soft-clDice on the boundary class. Topological-

lapsed the boundary predictions (mode-collapse:

model painted everything as field). Tried

on boundary class and field class, weight $\in \{0.2, 0.5\}$; abandoned.

- Curriculum boundary dilation ($3 \rightarrow 0$ px over training). Marginal vs the fixed-dilation baseline (-0.4 pts pixel IoU); the model is robust to boundary thickness on its own and the curriculum adds complexity without payoff.
- Frame-field auxiliary head [3] at our weight setting (0.1 alignment + 0.01 smoothness on top of SDF). No held-out gain; degrades the underlying SDF head's watershed signal (obj F1 collapses to the connected-components value). A wider weight sweep and target-rescaling under resize remain open.
- CutMix with 2px ignore buffer. Pasting a rectangular patch from another batch sample (with ignore-buffer along cut edges) produced -1 pt pixel IoU and near-zero change in object F1 vs augplus. Field-level structure does not transfer between patches the way object instances do.
- D4 test-time augmentation. The 8-way flip+rotation ensemble at inference was < 1 pt either way across all checkpoints we tried (sometimes negative), so we disabled it by default. The training augmentations already cover D4 symmetries.
- Mixed-country validation set. Adding cambodia + slovenia to the val set (instead of austria only) produced a noisier val/loss signal and shifted checkpoint selection away from the best held-out generalizer; held-out IoU dropped by ~ 4 pts at fixed recipe. Austria-only is the better monitor.
- Replicate padding instead of zero padding. We expected replicate-mode padding (`F.pad(mode="replicate")`) for image inputs at train and inference to be a free win by keeping BatchNorm inputs in distribution. In practice it consistently hurt object F1 by 8–10 pts on the Austria val set throughout training, while pixel IoU stayed within ± 0.02 of zero padding. We attribute this to replicate creating a synthetic extension of field interior into the padded region: predictions near the patch edge then learn that fields do not end abruptly, which causes adjacent fields to merge as a single connected component and collapse instance separation. Zero padding gives the model a consistent (if artificial) “everything is background here” boundary cue at the patch edge that keeps watershed clean.

7. Limitations and future work

Coverage. 5,715 (4.1%) of (patch, window) targets never produced a usable URL despite repeated reactivation; another 343 (0.2%) hit hard Planet-side asset failures. These patches still have their original FTW Sentinel-2 imagery but no PlanetScope companion in this release.

Temporal misalignment. Our PlanetScope acquisition date is the closest cloud-free match to the FTW season midpoint, not to the exact Sentinel-2 acquisition date. The gap is usually under two weeks but can exceed a month in cloudy regions.

Hard transfer regions. Portugal (terraced micro-fields) and Kenya (presence-only labels) collapse to ≤ 0.10 pixel IoU under every configuration we tried. These countries need either targeted data acquisition or a transfer-specific strategy. We do not try to solve them here.

Future work. (i) Rescale the SDF auxiliary target with `resize_aug` so the head doesn't degenerate under aggressive resizing. (ii) Train a B5 row to fill out the backbone $\times$ split grid. (iii) Add 8-band SuperDove samples where they are available, for richer spectral signal.

8. Conclusion

FTW-HD is a 3 m PlanetScope companion to Fields of The World. We did the boring engineering – range-read the COGs, reactivate the broken URLs, rasterize the labels onto the new grid – so that others don't have to. We also showed that a small B3 model trained with the right augmentations does as well as or better than the released $4\times$ larger Sentinel-2 baseline on every metric that does not bake in a pixel-grid bias. Field boundaries are where most of the loss happens, and a $3\times$ finer grid is the cheapest way to localize them.

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